

BlendFields: Few-Shot Example-Driven Facial Modeling

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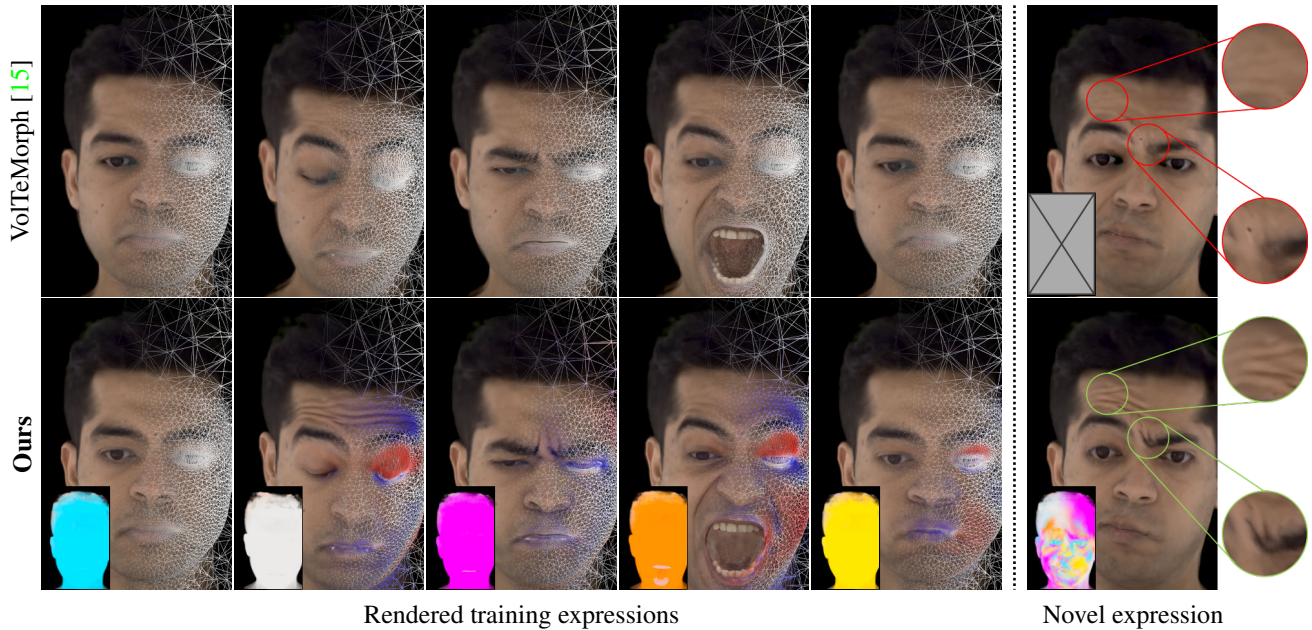


Figure 1. **Teaser** – Given five multi-view frames of different expressions, our approach generates a model capable of capturing the fine-grained details of a novel expression beyond the resolution of the underlying face model [15] (top right corner). This is achieved by *blending* the radiance fields computed for individual expressions, where the blending coefficients are modulated accordingly to *local* volumetric changes. These volumetric changes are measured as the difference in the tetrahedral volume of a mesh that deforms with the expression (■ increase, ■ decrease, and ■ no change in volume). Such an approach allows *BlendFields* to render sharp, expression-dependent details of the face without increasing the resolution of the mesh (bottom right corner).

Abstract

Generating faithful visualizations of human faces requires capturing both coarse and fine-level details of the face geometry and appearance. Existing methods are either data-driven, requiring an extensive corpus of data not publicly accessible to the research community, or fail to capture fine details because they rely on geometric face models that cannot represent fine-grained details in texture with a mesh discretization and linear deformation designed to model only a coarse face geometry. We introduce a method that bridges this gap by drawing inspiration from traditional computer graphics techniques. Unseen expressions

are modeled by blending appearance from a sparse set of extreme poses. This blending is performed by measuring local volumetric changes in those expressions and locally reproducing their appearance whenever a similar expression is performed at test time. We show that our method generalizes to unseen expressions, adding fine-grained effects on top of smooth volumetric deformations of a face, and demonstrate how it generalizes beyond faces.

1. Introduction

Recent advances in neural rendering of 3D scenes [53] offer 3D reconstructions of unprecedented quality [36] with an ever-increasing degree of control [23, 29]. Human faces are of particular interest to the research commu-

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	NeRF [36]	NeRFies [42]	HyperNeRF [43]	NeRFace [13]	NHA [17]	AVA [7]	VolTeMorph [15]	Ours
Applicability beyond faces	✓	✓	✓	✗	✗	✗	✓	✓
Interpretable control	✗	✗	✗	✓	✓	✗	✓	✓
Data efficiency	✗	✓	✓	✗	✓	✗	✓	✓
Expression-dependent changes	✗	✗	✓	✓	✓	✓	✗	✓
Generalizability to unknown expressions	✗	✗	✗	✓	✓	✗	✓	✓

Table 1. **Comparison** – We compare several methods to our approach. Other methods fall short in data efficiency and applicability. For example, AVA [7] requires 3.1 million training images while VolTeMorph [15] cannot model expression-dependent wrinkles realistically.

nity [1, 13–15] due to their application in creating realistic digital doubles [32, 53, 75, 79].

To render facial expressions not observed during training, current solutions [1, 13–15] rely on *parametric* face models [6]. These allow radiance fields [36] to be controlled by facial parameters estimated by off-the-shelf face trackers [27]. However, parametric models primarily capture smooth deformations and lead to digital doubles that lack realism because fine-grained and expression-dependent phenomena like wrinkles are not faithfully reproduced.

Authentic Volumetric Avatars (AVA) [7] overcomes this issue by learning from a large multi-view dataset of synchronized and calibrated images captured under controlled lighting. Their dataset covers a series of dynamic facial expressions and multiple subjects. However, this dataset remains unavailable to the public and is expensive to reproduce. Additionally, training models from such a large amount of data requires significant compute resources. To democratize digital face avatars, more efficient solutions in terms of hardware, data, and compute are necessary.

We address the efficiency concerns by building on the recent works in Neural Radiance Fields [15, 70, 74]. In particular, we extend VolTeMorph [15] to render facial details learned from images of a sparse set of expressions. To achieve this, we draw inspiration from blend-shape correctives [26], which are often used in computer graphics as a data-driven way to correct potential mismatches between a simplified model and the complex phenomena it aims to represent. In our setting, this mismatch is caused by the low-frequency deformations that the tetrahedral mesh from VolTeMorph [15], designed for real-time performance, can capture, and the high-frequency nature of expression wrinkles.

We train multiple radiance fields, one for each of the K sparse expressions present in the input data, and blend them to correct the low-frequency estimate provided by VolTeMorph [15]; see Fig. 1. We call our method *BlendFields* since it resembles the way blend shapes are employed in 3DMMs [6]. To keep K small (*i.e.*, to maintain a few-shot regime), we perform local blending to exploit the known correlation between wrinkles and changes in local differential properties [21, 45]. Using the dynamic geometry of [15], local changes in differential properties can be easily extracted by analyzing the tetrahedral representation underlying the corrective blendfields of our model.

Contributions. We outline the main qualitative differences between our approach and related works in Tab. 1, and our empirical evaluations confirm these advantages. In summary, we:

- extend VolTeMorph [15] to enable modeling of high-frequency information, such as expression wrinkles in a few-shot setting;
- introduce correctives [6] to neural field representations and activate them according to local deformations [45];
- make this topic more accessible with an alternative to techniques that are data and compute-intensive [7];
- show that our model generalizes beyond facial modeling, for example, in the modeling of wrinkles on a deformable object made of rubber.

2. Related Works

Neural Radiance Fields (NeRF) [36] is a method for generating 3D content from images taken with commodity cameras. It has prompted many follow-up works [4, 5, 19, 33, 35, 42, 46, 48, 52, 55, 68, 76] and a major change in the field for its photorealism. The main limitations of NeRF are its rendering speed, being constrained to static scenes, and lack of ways to control the scene. Rendering speed has been successfully addressed by multiple follow-up works [16, 18, 73]. Works solving the limitation to static scenes [1, 2, 22, 40, 59, 61, 67, 78] and adding explicit control [8, 23, 24, 50, 57, 72, 74] have limited resolution or require large amounts of training data because they rely on controllable coarse models of the scene (*e.g.*, 3DMM face model [6]) or a conditioning signal [43]. Methods built on an explicit model are more accessible because they require less training data but are limited by the model’s resolution. Our technique finds a sweet spot between these two regimes by using a limited amount of data to learn details missing in the controlled model and combining them together. Our experiments focus on faces because high-quality data and 3DMM face models are publicly available, which are the key component for creating digital humans.

2.1. Radiance Fields

Volumetric representations [56] have grown in popularity because they can represent complex geometries like hair more accurately than mesh-based ones. Neural Radiance Fields (NeRFs) [36] model a radiance volume with

a coordinate-based MLP learned from posed images. The MLP predicts density $\sigma(\mathbf{x})$ and color $\mathbf{c}(\mathbf{x}, \mathbf{v})$ for each point $\mathbf{x}$ in the volume and view direction $\mathbf{v}$ of a given camera. To supervise the radiance volume with the input images, each image pixel is associated with a ray $\mathbf{r}(t)$ cast from the camera center to the pixel, and samples along the ray are accumulated to determine the value of the image pixel $C(\mathbf{r})$:

$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t) \sigma(\mathbf{r}(t)) \mathbf{c}(\mathbf{r}(t), \mathbf{v}) dt, \quad (1)$$

where t_n and t_f are near and far planes, and

$$T(t) = \exp\left(-\int_{t_n}^t \sigma(\mathbf{r}(s)) ds\right), \quad (2)$$

is the transmittance function [51]. The weights of the MLP are optimized to minimize the mean squared reconstruction error between the target pixel and the output pixel. Several methods have shown that replacing the implicit functions approximated with an MLP for a function discretized on an explicit voxel grid results in a significant rendering and training speed-up [16, 18, 28, 49, 73].

2.2. Animating Radiance Fields

Several works exist to animate the scene represented as a NeRF. D-NeRF uses an implicit deformation model that maps sample positions back to a canonical space [44], but it cannot generalize to unseen deformations. Several works [13, 42, 43, 54] additionally account for changes in the observed scenes with a per-image latent code to model changes in color as well as shape, but it is unclear how to generalize the latents when animating a sequence without input images. Similarly, works focusing on faces [1, 13, 14, 80] use parameters of a face model to condition NeRF’s MLP, or learn a latent space of images and geometry [7, 30–32, 34, 58] that does not extrapolate beyond expressions seen during training.

In contrast to these approaches, we focus on using as little temporal training data as possible (*i.e.* five frames) while ensuring generalization. For this reason, we build our method on top of VolTeMorph [15], that uses a parametric model of the face to track the deformation of points in a volume around the face and builds a radiance field controlled by the parameters of a 3DMM. After training, the user can render an image for any expression of the face model. However, the approach cannot generate expression-dependent high-frequency details; see Fig. 1.

Similarly, NeRF-Editing [74] and NeRF Cages [70] propose to use tetrahedral meshes to deform a single-frame NeRF reconstruction. The resolution of the rendered scenes in these methods is limited by the resolution of the tetrahedral cage, which is constrained to a few thousand elements.

We discuss additional concurrent works in Supplementary.

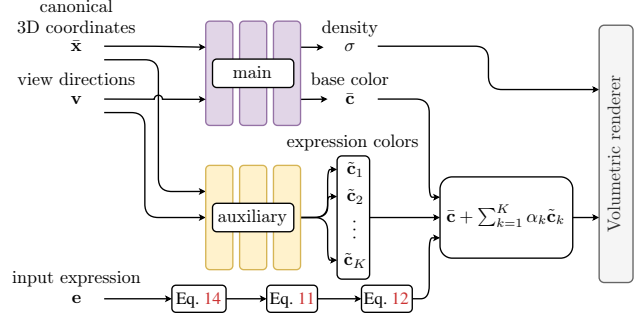


Figure 2. **BlendFields** – We implement our approach as a volumetric model, where the *appearance* (*i.e.* radiance) is the sum of the main appearance corrected by blending a small set of K expression-specific appearances. These appearances are learnt from extreme expressions, and then blended at test-time according to blend weights computed as a function of the input expression $\mathbf{e}$.

2.3. Tetrahedral Cages

To apply parametric mesh models, it is necessary to extend them to the volume to support the volumetric representation of NeRF. Tetrahedral cages are a common choice for their simplicity and ubiquity in computer graphics [15, 70, 72]. For example, VolTeMorph uses dense landmarks [63] to fit a parametric face model whose blend-shapes have been extended to a tetrahedral cage with finite elements method [9]. These cages can be quickly deformed and raytraced [37] using parallel computation on GPUs [11] while driving the volume into the target pose and allowing early ray termination for fast rendering. We further leverage the tetrahedral cage and use its differential properties [21], such as a local volume change, to model high-frequency details. For example, a change from one expression to another changes the volume of tetrahedra in regions where wrinkle formation takes place while it remains unchanged in flat areas. We can use this change in volume to select which of the trained NeRF expressions should be used for each tetrahedron to render high-frequency details.

3. Method

We introduce a volumetric model that can be driven by input expressions and visualize it in Fig. 2. We start this section by explaining our model and how we train and drive it with novel expressions utilizing parametric face models (Sec. 3.1). We then discuss how to compute measures of volume expansion and compression in the tetrahedra to combine volumetric models of different expressions (Sec. 3.2) and how we remove artifacts in out-of-distribution settings (Sec. 3.3). We conclude this section with implementation details (Sec. 3.4).

3.1. Our model

Given a neutral expression $\bar{\mathbf{e}}$, and a collection of posed images $\{C_c\}$ of this expression from multiple views, VolTe-

Morph [15] employs a map $\mathcal{T}$ to fetch the density and radiance¹ for a new expression $\mathbf{e}$ from the *canonical* frame defined by expression $\bar{\mathbf{e}}$:

$$\mathbf{c}(\mathbf{x}; \mathbf{e}) = \bar{\mathbf{c}}(\bar{\mathbf{x}}), \quad \bar{\mathbf{x}} = \mathcal{T}(\mathbf{x}; \mathbf{e} \rightarrow \bar{\mathbf{e}}) \quad (3)$$

$$\sigma(\mathbf{x}; \mathbf{e}) = \bar{\sigma}(\bar{\mathbf{x}}), \quad \bar{\mathbf{x}} = \mathcal{T}(\mathbf{x}; \mathbf{e} \rightarrow \bar{\mathbf{e}}) \quad (4)$$

$$\mathcal{L}_{\text{rgb}} = \mathbb{E}_{C \sim \{C_c\}} \mathbb{E}_{\mathbf{r} \sim C} \mathcal{L}_{\text{rgb}}^{\mathbf{r}} \quad (5)$$

$$\mathcal{L}_{\text{rgb}}^{\mathbf{r}} = \|C(\mathbf{r}; \mathbf{e}) - C(\mathbf{r})\|_2^2, \quad (6)$$

where $C(\mathbf{r}; \mathbf{e})$ is a pixel color produced by our model conditioned on the input expression $\mathbf{e}$, $C(\mathbf{r})$ is the ground-truth pixel color, and the mapping $\mathcal{T}$ is computed from smooth deformations of a tetrahedral mesh to render unseen expressions $\mathbf{e}$. We use expression vectors $\mathbf{e}$ from parametric face models, such as FLAME [27, 65]. However, as neither density nor radiance change with $\mathbf{e}$, changes in appearance are limited to the low-frequency deformations that $\mathcal{T}$ can express. For example, this model cannot capture high-frequency dynamic features like expression wrinkles. We overcome this limitation by conditioning radiance on expression. For this purpose, we assume radiance to be the sum of a template radiance (*i.e.* rest pose appearance of a subject) and K residual radiances (*i.e.* details belonging to corresponding facial expressions):

$$\mathbf{c}(\mathbf{x}; \mathbf{e}) = \bar{\mathbf{c}}(\mathbf{x}) + \sum_{k=1}^K \alpha_k(\mathbf{x}; \mathbf{e}) \cdot \tilde{\mathbf{c}}_k(\mathbf{x}), \quad (7)$$

We call our model *blend fields*, as it resembles the way in which blending is employed in 3D morphable models [6] or in wrinkle maps [41]. Note that we assume that pose-dependent geometry can be effectively modeled as a convex combination of colors $[\tilde{\mathbf{c}}(\mathbf{x})]_{k=1}^K$, since we employ the same density fields as in (4). In what follows, for convenience, we denote the vector field of blending coefficients as $\alpha(\mathbf{x}) = [\alpha_k(\mathbf{x})]_{k=1}^K$.

Training the model We train our model by assuming that we have access to a small set of K images $\{C_k\}$ (example in Fig. 3), each corresponding to an “extreme” expression $\{\mathbf{e}_k\}$, and minimize the loss:

$$\mathcal{L}_{\text{rgb}} = \mathbb{E}_k \mathbb{E}_{\mathbf{r}} \|C_K(\mathbf{r}; \mathbf{e}_k) - C_k(\mathbf{r})\|_2^2 \quad (8)$$

$$\text{where } \forall \mathbf{x}, \alpha(\mathbf{x}) = \mathbb{1}_k, \quad (9)$$

where $\mathbb{1}_k$ is the indicator vector, which has value one at the k -th position and zeroes elsewhere, and C_K represents the output of integrating the radiances in (7) along a ray.

Driving the model To control our model given a novel expression $\mathbf{e}$, we need to map the input expression code to the corresponding blendfield $\alpha(\mathbf{x})$. We parameterize the blend field as a vector field discretized on the vertices $\mathbf{V}(\mathbf{e})$ of

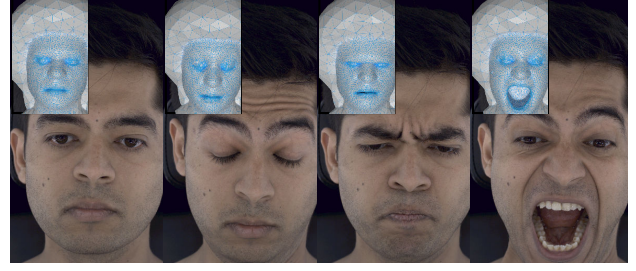


Figure 3. **Data** – We represent the data as a multi-view, multi-expression images. For each of these images, we obtain parameters of a parametric model, such as FLAME [27] to get: an expression vector $\mathbf{e}$ and a tetrahedral mesh described by vertices $\mathbf{V}(\mathbf{e})$. We highlight that our approach works for any object if a rough mesh and its descriptor are already provided.

our tetrahedral mesh, where the vertices deform according to the given expression. The field is discretized on vertices, but it can be queried within tetrahedra using linear FEM bases [38]. Our core intuition is that when the (local) geometry of the mesh matches the local geometry in one of the input expressions, the corresponding expression blend weight should be locally activated. More formally, let $\mathbf{v} \in \mathbf{V}$ be a vertex in the tetrahedra and $\mathcal{G}(\mathbf{v})$ a local measure of volume on the vertex described in Sec. 3.2, then

$$\mathcal{G}(\mathbf{v}(\mathbf{e})) \approx \mathcal{G}(\mathbf{v}(\mathbf{e}_k)) \implies \alpha(\mathbf{v}(\mathbf{e})) \approx \mathbb{1}_k. \quad (10)$$

To achieve this we first define a *local* similarity measure:

$$[\Delta \mathcal{G}_k(\mathbf{v}(\mathbf{e}))] = [\|\mathcal{G}(\mathbf{v}(\mathbf{e})) - \mathcal{G}(\mathbf{v}(\mathbf{e}_k))\|_2^2] \in \mathbb{R}^K \quad (11)$$

and then gate it with softmax (with temperature $\tau=10^6$) to obtain vertex blend weights:

$$\alpha(\mathbf{v}(\mathbf{e})) = \text{softmax}_{\tau} \{\Delta \mathcal{G}_k(\mathbf{v}(\mathbf{e}))\} \in [0, 1]^K \quad (12)$$

which realizes (10), as well as preserves the typically desirable characteristics of blend weights:

- *partition of unity*: $\forall \mathbf{x} \quad \alpha(\mathbf{x}) \in [0, 1]^K$ and $\|\alpha(\mathbf{x})\|_1 = 1$
- *activations sparsity*: minimizers of $\|\alpha(\mathbf{x})\|_0$

where the former ensures any reconstructed result is a *convex* combination of input data, and the latter prevents destructive interference [20].

3.2. Local geometry descriptor

Let us consider a tetrahedron as the matrix formed by its vertices $\mathbf{T} = \{\mathbf{v}_i\} \in \mathbb{R}^{3 \times 4}$, and its edge matrix as $\mathbf{D} = [\mathbf{v}_3 - \mathbf{v}_0, \mathbf{v}_2 - \mathbf{v}_0, \mathbf{v}_1 - \mathbf{v}_0]$. Let us denote $\bar{\mathbf{D}}$ as the edge matrix in rest pose and $\mathbf{D}$ as one of the deformed tetrahedra (*i.e.*, due to expression). From classical FEM literature, we can then compute the change in volume of the tetrahedra from the determinant of its deformation gradient [21]:

$$\Delta \mathcal{V}(\mathbf{T}) = \det(\mathbf{D} \cdot \bar{\mathbf{D}}^{-1}) \quad (13)$$

¹We omit view-dependent effects to simplify notation but include them in our implementation.

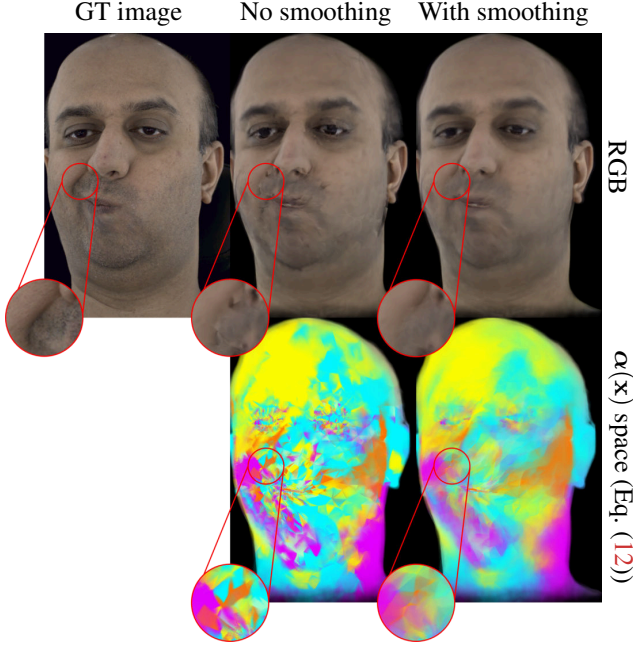


Figure 4. **Laplacian smoothing** – To combat artifacts stemming from calculating weights α across multiple expressions, which may assign different expressions to neighboring tetrahedra, we apply Laplacian smoothing [12]. As seen in the bottom row, smoothing gives a more consistent expression assignment.

We then build a local volumetric descriptor for a specific (deformed) vertex $\mathbf{v}(\mathbf{e})$ by concatenating the changes in volumes of neighboring (deformed) tetrahedra:

$$\mathcal{G}(\mathbf{v}(\mathbf{e})) = \bigoplus_{\mathbf{T} \in \mathcal{N}(\mathbf{v})} \Delta \mathcal{V}(\mathbf{T}(\mathbf{e})), \quad (14)$$

where $\bigoplus$ denotes concatenation and $\mathcal{N}(\mathbf{v})$ topological neighborhood of a vertex $\mathbf{v}$.

3.3. Blend-field smoothness

High-frequency spatial changes in blendfields can cause visual artifacts, see Fig. 4. We overcome this issue by applying a small amount of smoothing to the blendfield. Let us denote with $\mathbf{A} = \{\alpha(\mathbf{v}_v)\}$ the matrix of blend fields defined on all mesh vertices, and with $\mathbf{L}$ the Laplace-Beltrami operator for the tetrahedral mesh induced by linear bases [21]. We exploit the fact that at test-time, the field is discretized on the mesh vertices, execute a diffusion process on the tetrahedral manifold, and, to avoid instability problems, implement it via backward Euler [12]:

$$\mathbf{A}^{\text{diff}} = (\mathbf{I} - \lambda_{\text{diff}} \mathbf{L})^{-1} \mathbf{A}^n. \quad (15)$$

3.4. Implementation details

We build on VolTeMorph [15] and use its volumetric 3DMM face model. However, the same methodology can be used with other tetrahedral cages built on top of 3DMM

face models. The face model is created by extending the blendshapes of the parametric 3DMM face model [65] to a tetrahedral cage that defines the support in the neural radiance field. It has four bones controlling global rotation, the neck and the eyes with linear blend skinning, 224 expression blendshapes, and 256 identity blendshapes. Our face radiance fields are thus controlled and posed with the identity, expression, and pose parameters of the 3DMM face model [65], can be estimated by a real-time face tracking system like [64], and generalize convincingly to expressions representable by the face model.

Training. During training, we sample rays from a single frame to avoid out-of-memory issues when evaluating the tetrahedral mesh for multiple frames. Each batch contains 1024 rays. We sample $N_{\text{coarse}}=128$ points along a single ray during the coarse sampling and $N_{\text{importance}}=64$ for the importance sampling. We train the network to minimize the loss in Eq. (8) and sparsity losses with standard weights used in VolTeMorph [15, 18]. We train the methods for 5×10^5 steps using Adam [25] optimizer with learning rate 5×10^{-4} decaying exponentially by factor of 0.1 every 5×10^5 steps.

Inference. During inference, we leverage the underlying mesh to sample points around tetrahedra hit by a single ray. Therefore, we perform a single-stage sampling with $N=N_{\text{coarse}}+N_{\text{importance}}$ samples along the ray. When extracting the features (Eq. (14)), we consider $|\mathcal{N}(\mathbf{v})|=20$ neighbors. For the Laplacian smoothing, we set $\lambda_{\text{diff}}=0.1$ and perform a single iteration step. Geometric-related operations impose negligible computational overhead.

4. Experiments

We evaluate all methods on data of four subjects from the publicly available Multiface dataset [66]. We track the face for eight manually-selected "extreme" expressions. We then select $K=5$ expressions the combinations of which show as many wrinkles as possible. Each subject was captured with ≈ 38 cameras which gives ≈ 190 training images per subject². We use Peak Signal To Noise Ratio (PSNR) [3], Structural Similarity Index (SSIM) [60] and Learned Perceptual Image Patch Similarity (LPIPS) [77] to measure the performance of the models. Each of the rendered images has a resolution of 334×512 pixels.

As baselines, we use the following approaches: the original, static NeRF [36], NeRF conditioned on an expression code concatenated with input points $\mathbf{x}$, NeRFies [42], HyperNeRF³ [43], and VolTeMorph [15]. We replace the learnable code in NeRFies and HyperNeRF with the expression code $\mathbf{e}$ from the parametric model. Since VolTeMorph can be trained on multiple frames, which should

²To train BlendFields for a single subject we use $\approx 0.006\%$ of the dataset used by AVA [7].

³We use two architectures proposed by Park *et al.* [43].

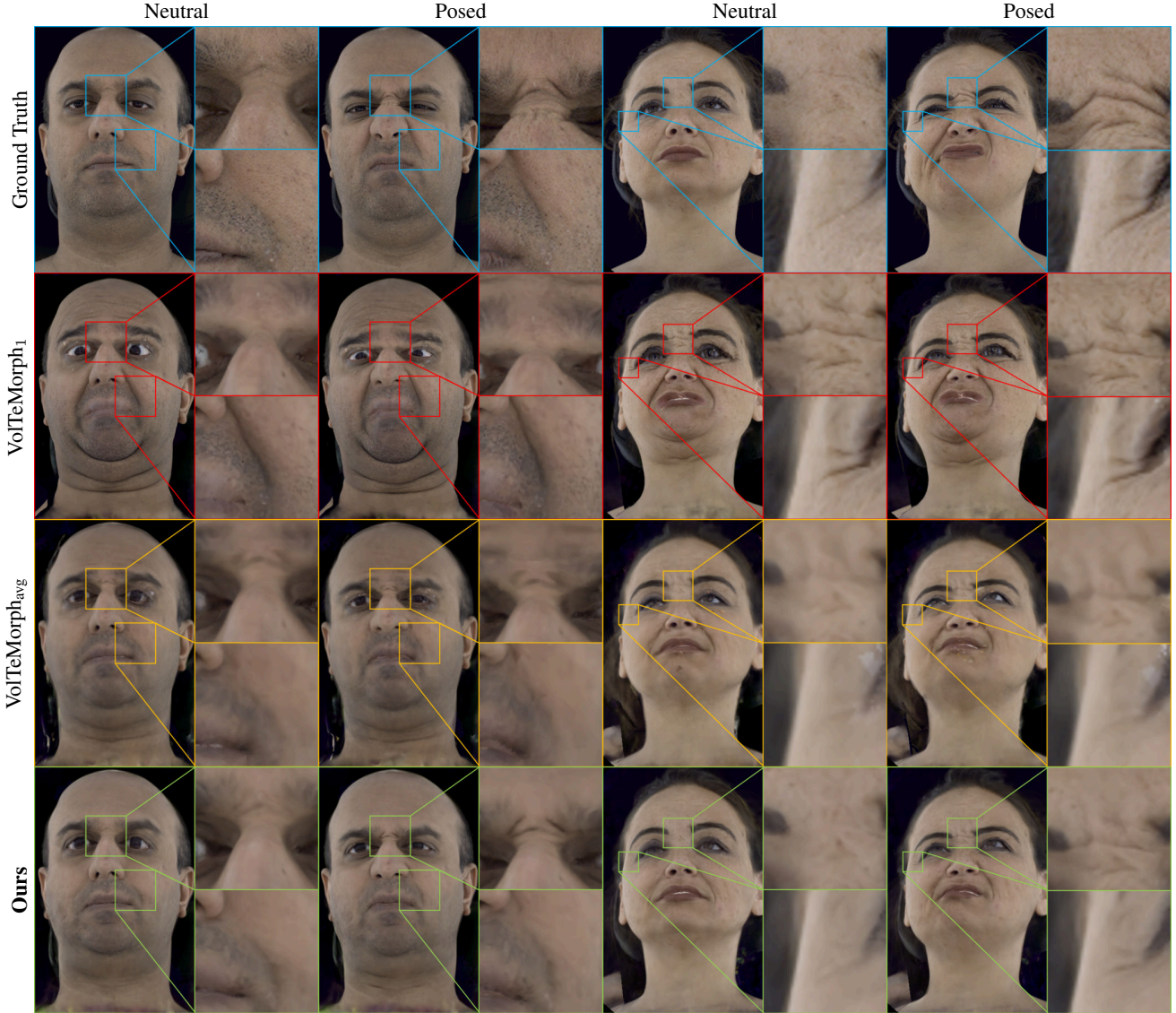


Figure 5. **Novel expression synthesis** – We compare qualitatively BlendFields with selected baselines (vertical) across two selected subjects (horizontal). Firstly, we show a neutral pose of the subject and then any of the available expressions. To our surprise, VolTeMorph_{avg} trained on multiple frames renders some details but with much lower fidelity. We argue that VolTeMorph_{arg} considers rendering wrinkles as artifacts that depend on the view direction (see Equation (1)). VolTeMorph₁ is limited to producing the wrinkles it was trained for. In contrast to those baselines, **BlendFields** captures the details and generalizes outside of the distribution. Please refer to the Supplementary for animated sequences and results for other methods.

lead to averaging of the output colors, we split it into two regimes: one trained on the most extreme expression⁴ (VolTeMorph₁) and the another trained on all available expressions (VolTeMorph_{avg})⁵. We use both of these baselines as VolTeMorph was originally designed for a single-frame scenario. By using two versions, we show that it is not trivial to extend it to multiple expressions.

⁴We manually select one frame that has the most visible wrinkles.

⁵We do not compare to NeRFace [13] and NHA [17] as VolTeMorph [15] performs better quantitatively than these methods.

4.1. Realistic Human Captures

Novel expression synthesis. We extract eight multi-view frames from the Multiface dataset [66], each of a different expression. Five of these expressions serve as training data, and the rest are used for evaluation. After training, we can extrapolate from the training expressions by modifying the expression vector $\mathbf{e}$. We use the remaining three expressions: moving mouth left and right, and puffing cheeks, to evaluate the capability of the models to reconstruct other expressions. In Fig. 5 we show that BlendFields is the only

Method	Real Data						Synthetic Data		
	Casual Expressions			Novel Pose Synthesis			Novel Pose Synthesis		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
NeRF [36]	23.6465	0.7384	0.2209	25.6696	0.8127	0.1861	13.7210	0.6868	0.3113
Conditioned NeRF [36]	22.9106	0.7162	0.2029	24.7283	0.7927	0.1682	19.5971	0.8138	0.1545
NeRFies [42]	22.6571	0.7105	0.2271	24.8376	0.7990	0.1884	19.3042	0.8081	0.1591
HyperNeRF-AP [43]	22.6219	0.7087	0.2236	24.7119	0.7931	0.1848	19.3557	0.8132	0.1563
HyperNeRF-DS [43]	22.9299	0.7182	0.2241	24.9909	0.8007	0.1860	19.4637	0.8159	0.1526
VolTeMorph ₁ [15]	24.9939	0.8358	0.1164	26.7526	0.8749	0.0954	26.7033	0.9500	0.0394
VolTeMorph _{avg} [15]	26.9209	0.8912	0.1105	28.6866	0.9176	0.0982	30.2107	0.9815	0.0387
BlendFields	27.5977	0.9056	0.0854	29.7372	0.9311	0.0782	32.2900	0.9882	0.0231

Table 2. **Quantitative results** – We compare BlendFields to other related approaches. We split the real data into two settings: one with casual expressions of subjects and the other with novel, static expressions. For the real data, we only compute metrics on the face region, which we separate using an off-the-shelf face segmentation network [62]. Please refer to the [Supplementary](#) for the results that include the background in the metrics as well. We average results across frames and subjects. VolTeMorph_{avg} [15] is trained on all frames, while VolTeMorph₁ is trained on a single frame. HyperNeRF-AP/-DS follows the design principles from Park *et al.* [43]. The best results are colored in and second best results in . BlendFields performs best in most of the datasets and metrics. Please note that HyperNeRF-AP/DS and NeRFies predict a dense deformation field designed for dense data. However, our input data consists of a few static frames only where the deformation field leads to severe overfitting.

Parameter	Real Data						Synthetic Data		
	Casual Expressions			Novel Pose Synthesis			Novel Pose Synthesis		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
$ \mathcal{N}(\mathbf{v}) = 1$	27.5620	0.9043	0.0893	29.7269	0.9306	0.0815	32.2371	0.9882	0.0234
$ \mathcal{N}(\mathbf{v}) = 5$	27.5880	0.9054	0.0864	29.7548	0.9312	0.0789	32.2900	0.9882	0.0231
$ \mathcal{N}(\mathbf{v}) = 10$	27.5933	0.9054	0.0859	29.7456	0.9312	0.0785	32.3324	0.9882	0.0230
$ \mathcal{N}(\mathbf{v}) = 20$	27.5977	0.9056	0.0854	29.7372	0.9311	0.0782	32.7949	0.9887	0.0221
Without smoothing	27.2535	0.8959	0.0939	29.3726	0.9233	0.0846	32.2452	0.9876	0.0238
With smoothing	27.5977	0.9056	0.0854	29.7372	0.9311	0.0782	32.6323	0.9887	0.0223

Table 3. **Ablation study** – First, we check the effect of the neighborhood size $|\mathcal{N}(\mathbf{v})|$ on the results. Below that, we compare the effect of smoothing. The best results are colored in and the second best in . For the real dataset, changing the neighborhood size gives inconsistent results, while smoothing improves the rendering quality. In the synthetic scenario, setting $|\mathcal{N}(\mathbf{v})|=20$ and the Laplacian smoothing consistently gives the best results. The discrepancy between real and synthetic datasets is caused by inaccurate face tracking for the former. We describe this issue in detail in Section 4.4.

method capable of rendering convincing wrinkles dynamically, depending on the input expression. BlendFields performs favorably compared to the baselines (see Tab. 2).

Casual expressions. The Multiface dataset contains sequences where the subject follows a script of expressions to show during the capture. Each of these captures contains between 1000 and 2000 frames. This experiment tests whether a model can interpolate between the training expressions smoothly and generalize beyond the training data. Quantitative results are shown in Tab. 2. Our approach performs best all the settings. See animations in the [Supplementary](#) for a comparison of rendered frames across all methods.

4.2. Modeling Objects Beyond Faces

We show that our method can be applied beyond face modeling. We prepare two datasets containing 96 views per frame of bending and twisting cylinders made of a rubber-like material (24 and 72 temporal frames, respectively). When bent or twisted, the cylinders reveal pose-dependent details. The expression vector $\mathbf{e}$ now encodes time: 0 if the cylinder is in the canonical pose, 1 if it is posed, and any values between $[0, 1]$ for the transitioning stage. We select expressions $\{0, 0.5, 1.0\}$ as a training set (for VolTeMorph₁ we use 1.0 only). For evaluation, we take every fourth frame from the full sequence using cameras from the bottom and both sides of the object. We take the

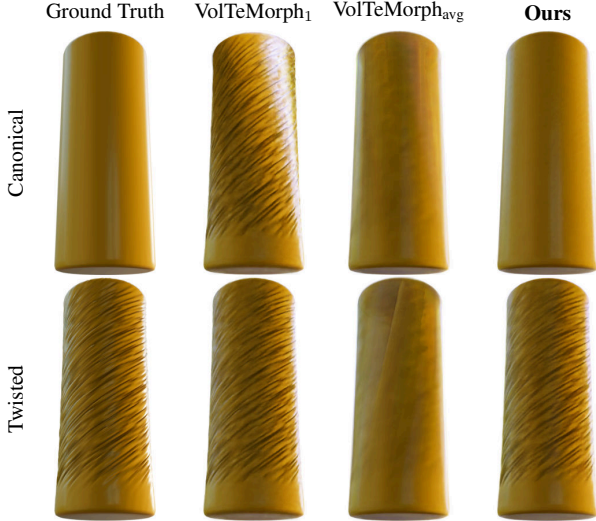


Figure 6. **Qualitative results on synthetic dataset** – For a simple dataset, baselines cannot model high-frequency, pose-dependent details. VolTeMorph₁ renders wrinkles for the straight pose as well, as it is trained for the twisted cylinder only, while VolTeMorph_{avg} averages out the texture.

mesh directly from Houdini [69], which we use for wrinkle simulation, and render the images in Blender [10]. We show quantitative results in Tab. 2 for the bending cylinder, and a comparison of the inferred images in Fig. 6 for the twisted one⁶. BlendFields accurately captures the transition from the rest configuration to the deformed state of the cylinder, rendering high-frequency details where required. All other approaches struggle with interpolation between states. VolTeMorph₁ (trained on a single extreme pose) renders wrinkles even when the cylinder is not twisted.

4.3. Ablations

We check how the neighborhood size $|\mathcal{N}(\mathbf{v})|$ and the application of the smoothing influence the performance of our method. We show the results in Tab. 3. BlendFields works best in most cases when considering a relatively wide neighborhood for the tetrahedral features⁷. Laplacian smoothing consistently improves the quality across all the datasets (see Fig. 4). We additionally present in the Supplementary how the number of expressions used for training affects the results.

4.4. Failure Cases

While BlendFields offers significant advantages for rendering realistic and dynamic high-frequency details, it falls short in some scenarios (see Fig. 7). One of the issues arises

⁶Our motivation is that it is easier to show pose-dependent deformations on twisting as it affects the object globally, while the bending cannot be modeled by all the baselines due to the non-stationary effects.

⁷Larger neighborhood sizes caused out-of-memory errors on our NVIDIA 2080Ti GPU.

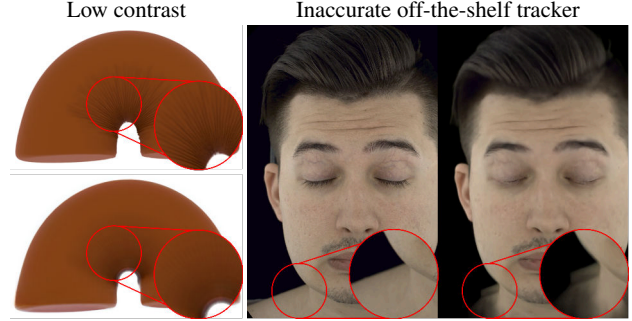


Figure 7. **Failure cases** – We show failure cases for our proposed approach. *Left*: In the presence of wrinkles in low-contrast images, BlendFields takes longer to converge to make wrinkles visible. We show the ground truth on the top, and rendering after training 7×10^5 steps on the bottom. In contrast, we rendered images in Figure 6 after 2×10^5 steps. *Right*: BlendFields inherits issues from VolTeMorph [15], which relies on the initial fit of the face mesh. If the fit is inaccurate, artifacts appear in the final render.

when the contrast between wrinkles and the subject’s skin color is low. In those instances, we observe a much longer time to convergence. Moreover, as we build BlendFields on VolTeMorph, we also inherit some of its problems. Namely, the method heavily relies on the initial fit of the parametric model – any inaccuracy leads to ghosting artifacts or details on the face that jump between frames.

5. Conclusions

We present a general approach, BlendFields, for rendering high-frequency expression-dependent details using NeRFs. BlendFields draws inspiration from classical computer graphics by blending expressions from the training data to render expressions unseen during training. We show that BlendFields renders images in a controllable and interpretable manner for novel expressions and can be applied to render human avatars learned from publicly available datasets. We additionally discuss the potential misuse of our work in the Supplementary.

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